

If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

- a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.
 - b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.
 - c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.
 - d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).
 - e. Bonus problems are not required for credit, but they are worth extra credit.
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Problems

Problem 1. (Logistic Function Properties.) The logistic (sigmoid) function is

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

- a. Show that $0 < \sigma(z) < 1$ for all real z .
- b. Show that $\sigma(-z) = 1 - \sigma(z)$.
- c. Compute $\sigma'(z)$ and show that

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)).$$

Solution.

- a. **Showing that** $0 < \sigma(z) < 1$ **for all real** z . Since $e^{-z} > 0$, the denominator $1 + e^{-z}$ is > 1 , so $0 < \sigma(z) < 1$.
- b. **Showing that** $\sigma(-z) = 1 - \sigma(z)$.

$$\sigma(-z) = \frac{1}{1 + e^z} = \frac{e^{-z}}{e^{-z} + 1} = 1 - \frac{1}{1 + e^{-z}} = 1 - \sigma(z).$$

- c. **Computing** $\sigma'(z)$. Taking the derivative of $\sigma(z)$, we have that

$$\sigma'(z) = \frac{e^{-z}}{(1 + e^{-z})^2} = \left(\frac{1}{1 + e^{-z}} \right) \left(\frac{e^{-z}}{1 + e^{-z}} \right) = \sigma(z)(1 - \sigma(z)).$$

Problem 2. (Binary Logistic Regression.) Consider binary logistic regression with data $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and $y_i \in \{0, 1\}$. The logistic model is

$$p(y = 1 \mid x; \theta) = \sigma(\theta^\top x), \quad \text{where } \sigma(z) = \frac{1}{1 + e^{-z}}.$$

The (unregularized) negative log-likelihood is

$$\ell(\theta) = - \sum_{i=1}^n \left[y_i \log(\sigma(\theta^\top x_i)) + (1 - y_i) \log(1 - \sigma(\theta^\top x_i)) \right].$$

- Derive an expression for the gradient $\nabla_\theta \ell(\theta)$.
- Derive an expression for the Hessian $\nabla^2 \ell(\theta)$.
- Show that $\ell(\theta)$ is convex.

Hint: you may want to consider one point i.e. $n = 1$, and then generalize. **Solution.** Let $z_i = \theta^\top x_i$ and $p_i = \sigma(z_i)$.

- Gradient.** For a single example (x_i, y_i) , the contribution to ℓ is

$$\ell_i(\theta) = - \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right].$$

Differentiate with respect to θ using the chain rule to get that

$$\frac{\partial \ell_i}{\partial \theta} = \frac{\partial \ell_i}{\partial z_i} \frac{\partial z_i}{\partial \theta}.$$

First,

$$\frac{\partial \ell_i}{\partial z_i} = - \left[y_i \frac{1}{p_i} \frac{\partial p_i}{\partial z_i} + (1 - y_i) \frac{1}{1 - p_i} (-1) \frac{\partial p_i}{\partial z_i} \right].$$

We know $\frac{\partial p_i}{\partial z_i} = p_i(1 - p_i)$, so

$$\frac{\partial \ell_i}{\partial z_i} = - \left[y_i \frac{1}{p_i} p_i(1 - p_i) - (1 - y_i) \frac{1}{1 - p_i} p_i(1 - p_i) \right] = - [y_i(1 - p_i) - (1 - y_i)p_i].$$

Simplify:

$$y_i(1 - p_i) - (1 - y_i)p_i = y_i - y_i p_i - p_i + y_i p_i = y_i - p_i.$$

Therefore,

$$\frac{\partial \ell_i}{\partial z_i} = p_i - y_i.$$

Next, $\frac{\partial z_i}{\partial \theta} = x_i$, so

$$\frac{\partial \ell_i}{\partial \theta} = (p_i - y_i)x_i.$$

Summing over i ,

$$\nabla_\theta \ell(\theta) = \sum_{i=1}^n (p_i - y_i)x_i = \sum_{i=1}^n (\sigma(\theta^\top x_i) - y_i)x_i.$$

In matrix form, if $X \in \mathbb{R}^{n \times d}$ has rows $x_i^\top$, and $p = (\sigma(\theta^\top x_1), \dots, \sigma(\theta^\top x_n))^\top$, $y = (y_1, \dots, y_n)^\top$, then

$$\nabla_\theta \ell(\theta) = X^\top (p - y).$$

b. **Hessian.** Differentiate the gradient again to get that

$$\nabla_{\theta}^2 \ell(\theta) = \sum_{i=1}^n \frac{\partial}{\partial \theta} \left[(p_i - y_i) x_i \right] = \sum_{i=1}^n \left(\frac{\partial p_i}{\partial \theta} x_i^{\top} \right),$$

since y_i and x_i are constants. We have

$$\frac{\partial p_i}{\partial \theta} = \frac{\partial p_i}{\partial z_i} \frac{\partial z_i}{\partial \theta} = p_i(1 - p_i) x_i.$$

Thus,

$$\nabla_{\theta}^2 \ell(\theta) = \sum_{i=1}^n p_i(1 - p_i) x_i x_i^{\top}.$$

In matrix form, define a diagonal matrix

$$W = \text{diag}(p_1(1 - p_1), \dots, p_n(1 - p_n)).$$

Then

$$\nabla_{\theta}^2 \ell(\theta) = X^{\top} W X.$$

c. **Convexity.** To show convexity, it is enough to show the Hessian is positive semidefinite (PSD). For any $v \in \mathbb{R}^d$,

$$v^{\top} \nabla_{\theta}^2 \ell(\theta) v = v^{\top} \left(\sum_{i=1}^n p_i(1 - p_i) x_i x_i^{\top} \right) v = \sum_{i=1}^n p_i(1 - p_i) (v^{\top} x_i)^2.$$

Since $p_i(1 - p_i) \geq 0$ and $(v^{\top} x_i)^2 \geq 0$, each term is nonnegative, so the sum is nonnegative:

$$v^{\top} \nabla_{\theta}^2 \ell(\theta) v \geq 0 \quad \forall v.$$

Hence $\nabla_{\theta}^2 \ell(\theta)$ is PSD and $\ell(\theta)$ is convex.

Problem 3. (Regularized Logistic Regression.) Consider ℓ_2 -regularized logistic regression problem:

$$\min_{\theta \in \mathbb{R}^d} J(\theta) = \ell(\theta) + \frac{\lambda}{2} \|\theta\|_2^2,$$

where $\ell(\theta)$ is the negative log-likelihood from **Problem 2** and $\lambda \geq 0$.

- Derive the gradient $\nabla_{\theta} J(\theta)$.
- Explain qualitatively what happens to the learned parameter θ as $\lambda \rightarrow 0$ and as $\lambda \rightarrow \infty$.

Solution.

a. **Gradient of the regularized objective.** We have

$$J(\theta) = \ell(\theta) + \frac{\lambda}{2} \|\theta\|_2^2.$$

From **Problem 1**, we have that

$$\nabla_{\theta} \ell(\theta) = \sum_{i=1}^n (\sigma(\theta^{\top} x_i) - y_i) x_i.$$

The gradient of the regularization term is

$$\nabla_{\theta} \left(\frac{\lambda}{2} \|\theta\|_2^2 \right) = \lambda \theta.$$

Thus,

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \ell(\theta) + \lambda \theta = \sum_{i=1}^n (\sigma(\theta^{\top} x_i) - y_i) x_i + \lambda \theta.$$

b. **Behavior as $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.**

- As $\lambda \rightarrow 0$: The regularization term vanishes, and we recover the unregularized maximum likelihood estimate. If the data are linearly separable, $\|\theta\|$ can grow very large (the logistic loss can push the margin to infinity), leading to extremely confident predictions and potential overfitting.
- As $\lambda \rightarrow \infty$: The regularization term dominates, strongly penalizing large $\|\theta\|$. In the limit, the minimizer tends toward $\theta \approx 0$, so $p(y = 1 \mid x) \approx \sigma(0) = 1/2$ for all x . The model becomes almost constant and underfits the data.

Problem 4. (Decision Boundary in Logistic Regression.) Consider logistic regression in $\mathbb{R}^2$ with parameters $\theta = (\theta_0, \theta_1, \theta_2)$. The model is

$$p(y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2),$$

where $x = (x_1, x_2)$.

- Show that the decision boundary defined by $p(y = 1 \mid x) = 0.5$ is a straight line in the (x_1, x_2) -plane.
- Write the equation of this line explicitly in terms of $\theta_0, \theta_1, \theta_2$.
- Suppose we augment the feature vector with a quadratic feature x_1^2 , i.e., consider the model

$$p(y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2).$$

Describe qualitatively the shape of the decision boundary now.

Solution.

- Linearity of the decision boundary.** The decision boundary $p(y = 1 \mid x) = 0.5$ is obtained by solving

$$\sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2) = 0.5.$$

Recall that $\sigma(z) = 1/(1 + e^{-z})$ and $\sigma(z) = 0.5$ if and only if $z = 0$. Hence,

$$\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0.$$

This is the equation of an affine hyperplane in $\mathbb{R}^2$, i.e., a straight line.

b. **Equation of the line.** Solving for x_2 (assuming $\theta_2 \neq 0$),

$$x_2 = -\frac{\theta_1}{\theta_2}x_1 - \frac{\theta_0}{\theta_2}.$$

So the decision boundary is the line with slope $-\theta_1/\theta_2$ and intercept $-\theta_0/\theta_2$.

c. **Adding a quadratic feature.** Now consider

$$p(y = 1 | x) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2).$$

The decision boundary is given by the set of points (x_1, x_2) satisfying

$$\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 = 0.$$

For fixed θ , this is a quadratic curve in the plane (a conic section). For example, if $\theta_3 \neq 0$ and $\theta_2 \neq 0$, we can solve for x_2 :

$$x_2 = -\frac{\theta_3}{\theta_2}x_1^2 - \frac{\theta_1}{\theta_2}x_1 - \frac{\theta_0}{\theta_2}.$$

This is a parabola in the (x_1, x_2) -plane. More generally, augmenting features with higher-order or interaction terms makes the decision boundary nonlinear in the original input space, even though the model remains linear in the parameters.

Problem 5. (A Mathematical View of Bias and Variance.) Let the true model be linear:

$$y = w^{\star\top} x,$$

where $x \in \mathbb{R}^d$ is fixed and known. Suppose we collect n training samples

$$y_i = w^{\star\top} x_i + \varepsilon_i,$$

where the noise terms satisfy

$$\mathbb{E}[\varepsilon_i] = 0, \quad \mathbb{E}[\varepsilon_i \varepsilon_j] = \sigma^2 \delta_{ij} = \begin{cases} 0, & i \neq j \\ \sigma^2, & i = j \end{cases}.$$

Note that $\mathbb{E}[\cdot]$ is linear, i.e., $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. Let $X \in \mathbb{R}^{n \times d}$ have rows $x_i^\top$ and $y = (y_1, \dots, y_n)^\top$. Consider two estimators:

$$\hat{w}_{\text{LS}} = (X^\top X)^{-1} X^\top y, \quad \hat{w}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y,$$

where $\lambda \geq 0$.

a. Compute the **bias** of $\hat{w}_\lambda$, defined as

$$\text{Bias}(\hat{w}_\lambda) = \mathbb{E}[\hat{w}_\lambda] - w^\star.$$

b. Compute the **covariance** (or variance matrix)

$$\text{Var}(\hat{w}_\lambda) = \mathbb{E}[(\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])(\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])^\top].$$

c. Show that the **mean squared prediction error** at a test point x decomposes as

$$\mathbb{E}[(x^\top \hat{w}_\lambda - x^\top w^*)^2] = \underbrace{x^\top \text{Var}(\hat{w}_\lambda) x}_{\text{variance term}} + \underbrace{(x^\top \text{Bias}(\hat{w}_\lambda))^2}_{\text{bias}^2}.$$

d. Using the above formulas, explain mathematically why increasing λ

- increases bias, and
- decreases variance.

Solution.

a. **Bias of the ridge estimator.** Let $A = X^\top X$.

We have

$$\hat{w}_\lambda = (A + \lambda I)^{-1} X^\top (X w^* + \varepsilon) = (A + \lambda I)^{-1} (A w^* + X^\top \varepsilon).$$

Taking expectation and using $\mathbb{E}[\varepsilon] = 0$:

$$\mathbb{E}[\hat{w}_\lambda] = (A + \lambda I)^{-1} A w^*.$$

Thus the bias is

$$\text{Bias}(\hat{w}_\lambda) = \mathbb{E}[\hat{w}_\lambda] - w^* = ((A + \lambda I)^{-1} A - I) w^*.$$

Multiply numerator and denominator inside by $(A + \lambda I)$:

$$(A + \lambda I)^{-1} A - I = (A + \lambda I)^{-1} (A - (A + \lambda I)) = -\lambda (A + \lambda I)^{-1}.$$

Hence

$$\boxed{\text{Bias}(\hat{w}_\lambda) = -\lambda (A + \lambda I)^{-1} w^*}.$$

b. **Variance of the ridge estimator.** Just in case, the reason $\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda] = (X^\top X + \lambda I)^{-1} X^\top \varepsilon$ is because we plug in $(w^*)^\top x + \varepsilon$ for a new random sample like in part a. Then we have

$$\hat{w}_\lambda = (X^\top X + \lambda I)^{-1} X^\top ((w^*)^\top x + \varepsilon)$$

and

$$\mathbb{E}[\hat{w}_\lambda] = (X^\top X + \lambda I)^{-1} X^\top (w^*)^\top x$$

because $\mathbb{E}[\varepsilon] = 0$. Thus

$$\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda] = (X^\top X + \lambda I)^{-1} X^\top \varepsilon$$

We have

$$\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda] = (A + \lambda I)^{-1} X^\top \varepsilon.$$

Thus,

$$\text{Var}(\hat{w}_\lambda) = (A + \lambda I)^{-1} X^\top \mathbb{E}[\varepsilon \varepsilon^\top] X (A + \lambda I)^{-1}.$$

Since $\mathbb{E}[\varepsilon \varepsilon^\top] = \sigma^2 I$,

$$\boxed{\text{Var}(\hat{w}_\lambda) = \sigma^2 (A + \lambda I)^{-1} A (A + \lambda I)^{-1}}.$$

c. **Bias–variance decomposition of prediction error.**

For a fixed x , we have that

$$x^\top \hat{w}_\lambda - x^\top w^\star = x^\top (\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda]) + x^\top (\mathbb{E}[\hat{w}_\lambda] - w^\star).$$

so that

$$(x^\top \hat{w}_\lambda - x^\top w^\star)^2 = (x^\top (\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda]))^2 + 2x^\top (\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])x^\top (\mathbb{E}[\hat{w}_\lambda] - w^\star) + (x^\top (\mathbb{E}[\hat{w}_\lambda] - w^\star))^2.$$

Then

$$\mathbb{E}[2x^\top (\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])x^\top (\mathbb{E}[\hat{w}_\lambda] - w^\star)] = 0$$

since $\mathbb{E}[\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda]] = \mathbb{E}[\hat{w}_\lambda] - \mathbb{E}[\hat{w}_\lambda] = 0$.

The first term can be written as

$$\begin{aligned} (x^\top (\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda]))^2 &= x^\top (\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])x^\top (\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])^\top x \\ &= x^\top (\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])(\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])^\top x \\ &= x^\top (\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])(\hat{w}_\lambda - \mathbb{E}[\hat{w}_\lambda])^\top x. \end{aligned}$$

Thus

$$\mathbb{E}[(x^\top \hat{w}_\lambda - x^\top w^\star)^2] = x^\top \text{Var}(\hat{w}_\lambda)x + (x^\top \text{Bias}(\hat{w}_\lambda))^2.$$

d. **Effect of λ .**

Use the closed-form expressions to get that

$$\text{Bias}(\hat{w}_\lambda) = -\lambda(A + \lambda I)^{-1}w^\star.$$

As λ increases, we have that

- $(A + \lambda I)^{-1}$ decreases in magnitude,
- but multiplied by λ , the overall term grows.

Thus $\|\text{Bias}\|$ increases with λ . This makes sense because as λ increases, the model becomes more regularized, the variance of the parameters decreases, and the learned model is focusing more on optimizing the training loss term in the objective.

The variance is given by

$$\text{Var}(\hat{w}_\lambda) = \sigma^2(A + \lambda I)^{-1}A(A + \lambda I)^{-1}.$$

As λ increases, $(A + \lambda I)^{-1}$ shrinks, so the entire matrix shrinks. Thus $\text{Var}(\hat{w}_\lambda)$ decreases in magnitude.

To summarize, we have that

$$\lambda \uparrow \implies \text{bias} \uparrow, \text{variance} \downarrow.$$

Problem 6. (Interpreting Logistic Regression Coefficients in 1D.) Consider 1D logistic regression with feature $x \in \mathbb{R}$:

$$p(y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x),$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$. That is the probability of $y = 1$ given x , is given by the sigmoid function. The log-odds is given by the logit function, and in the context of logistic regression, the *odds* that $y = 1$ given the feature x is

$$\text{odds}(x) = \frac{p(y = 1 \mid x)}{1 - p(y = 1 \mid x)}.$$

- a. Show that the *log-odds* (logit) is affine in x : that is, show that

$$\log \left(\frac{p(y = 1 \mid x)}{1 - p(y = 1 \mid x)} \right) = \theta_0 + \theta_1 x.$$

- b. Suppose $\theta_1 > 0$. Show that increasing x by Δ multiplies the odds by a factor of $e^{\theta_1 \Delta}$.
 c. If $\theta_1 = 0.7$, what is the factor by which the odds change when x increases by 2 units?

Solution.

- a. Let $p = p(y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x)$. Then

$$p = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}} \implies \frac{p}{1 - p} = e^{\theta_0 + \theta_1 x}.$$

Taking log of both sides, we get that

$$\log \left(\frac{p}{1 - p} \right) = \theta_0 + \theta_1 x.$$

- b. For x and $x + \Delta$, the log-odds differ by

$$(\theta_0 + \theta_1(x + \Delta)) - (\theta_0 + \theta_1 x) = \theta_1 \Delta.$$

Thus the odds are multiplied by $e^{\theta_1 \Delta}$:

$$\frac{p(y = 1 \mid x + \Delta)/(1 - p(y = 1 \mid x + \Delta))}{p(y = 1 \mid x)/(1 - p(y = 1 \mid x))} = e^{\theta_1 \Delta}.$$

- c. If $\theta_1 = 0.7$ and $\Delta = 2$, then the odds are multiplied by

$$e^{\theta_1 \Delta} = e^{0.7 \cdot 2} = e^{1.4}.$$